

Analysis Note: $\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau)$ with an AI-Agent-Driven, Open-Source Fast-Simulation Framework

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with an LLM analysis agent (Claude) operating under human review

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This study uses only publicly available tools and publications. No Belle II internal materials — data, MC, internal notes, or the software framework — are used. Everything is open-source and runs on a laptop.

Abstract

We are developing a framework in which an LLM agent carries out branching-fraction analyses under explicit human review: the agent plans, a human (or a second, checklist-driven LLM reviewer) approves, the agent executes typed analysis tools — EvtGen and Pythia8 generation, a parametric fast detector simulation, `uproot` ntuples, `pyhf` fits — and finishes by writing the analysis note. This document, which follows the structure of the notes the agent itself produces, records a complete demonstration: a measurement of $\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau)$ on a luminosity-matched pseudo-dataset of 10^6 generic $B\bar{B}$ plus 2.4×10^6 continuum events ($\approx 0.9 \text{ fb}^{-1}$ equivalent). A binned m_{miss}^2 template fit with simultaneous electron and muon channels returns $\mathcal{B} = (1.48 \pm 0.69)\%$ against a generator truth of 1.48%: the framework recovers the input with correctly estimated uncertainties (validated with toy experiments), corresponding to $\sim 1.4\%$ relative statistical precision at 1 ab^{-1} . The purpose of this note is to show colleagues concretely what this approach produces end to end.

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1 Introduction

LLM-based agents have recently been shown to carry out complete high-energy-physics analyses autonomously on ALEPH, DELPHI, and CMS open data, from event selection through paper drafting [1]. Our interest is narrower and, we believe, closer to how such tools could enter a collaboration’s workflow: *branching-fraction measurements*, a class of analyses with a well-defined and repeatable structure, executed by an agent under a mandatory review gate, with a cut-based selection policy in which every step can be reproduced by hand.

The agent’s workflow is Plan → review → Execute → Report → Note. At the Plan stage the agent must state the samples (with event counts and random seeds), the selection strategy, the published analysis whose variable choices it follows, and the deliverables; the reviewer — a human at the terminal, or optionally a second LLM instance evaluating a fixed checklist — approves or returns actionable feedback, upon which the agent replans. Every number in the final note is required to originate from a tool result of the session; the model performs no arithmetic of its own.

As the demonstration channel we use $B^0 \rightarrow D^{*-}\tau^+\nu_\tau$ with $B^0 \rightarrow D^{*-}\ell^+\nu_\ell$ ($\ell = e, \mu$) as the normalization mode — kinematics familiar from the $R(D^*)$ program. The discriminating variables follow the $B \rightarrow D^*\ell\nu$ and $R(D^*)$ analyses of BaBar, Belle, and Belle II [2, 3, 4, 5, 6]: the D^0 -mass and Δm windows, the Fox–Wolfram ratio R_2 against continuum, an extra-energy-type requirement on the rest of event, and the missing mass squared m_{miss}^2 as the τ/ℓ discriminant. The framework itself (public repository github.com/sykeisuke/phys_agent) is mode-agnostic.

2 Samples

2.1 Event generation

All samples are generated on a laptop. $\Upsilon(4S)$ mesons are produced with a Belle II-like boost ($E_{e^-} = 7\text{ GeV}$, $E_{e^+} = 4\text{ GeV}$ head-on, $p_z \simeq 3\text{ GeV}$) and decayed with EvtGen 2.2.3 [7]; final-state radiation is generated with Photos. $B^0\bar{B}^0$ pairs are produced with the standard VSS_BMIX model (coherent mixing). For the forced signal and control samples one B is aliased to the chain of interest while the other decays generically according to the DECAY.DEC tables shipped with EvtGen; τ decays use EvtGen’s native models (TAULNUNU etc.) rather than Tauola, whose conda-forge `osx-arm64` build is unusable. The forced chains are fully charged where possible:

- signal: $B^0 \rightarrow D^{*-}\tau^+\nu_\tau$ (ISGW2 [8]), $D^{*-} \rightarrow \bar{D}^0\pi^-$, $\bar{D}^0 \rightarrow K^+\pi^-$, $\tau^+ \rightarrow \ell^+\nu\bar{\nu}$ ($\ell = e, \mu$ in the τ -table ratio);

- normalization: $B^0 \rightarrow D^{*-}\ell^+\nu$ (ISGW2), same D^* chain;
- feed-down: $B^0 \rightarrow D^{**}\ell\nu$ with the narrow states $D_1(2420)$ and $D_2^*(2460)$ forced to $D^{*-}\pi^0$;
- generic $\Upsilon(4S) \rightarrow B\bar{B}$ (nothing forced; `DECAY.DEC`);
- continuum $e^+e^- \rightarrow q\bar{q}$ ($q = u, d, s, c$) with Pythia8 [9], identical beam configuration, processed by the same detector response in a single pass.

2.2 Sample list and luminosity equivalence

Table 1: Monte Carlo samples. All are reproducible from the listed seeds with the commands in the repository README.

Process	Model	Events	Seeds	Role
$B^0 \rightarrow D^{*-}\tau^+\nu, \tau \rightarrow \ell\nu\nu$	ISGW2	5 000	1/42	signal efficiency
$B^0 \rightarrow D^{*-}\ell^+\nu$	ISGW2	5 000	2/43	normalization, CR checks
$B^0 \rightarrow D^{**}\ell\nu$	ISGW2	5 000	3/44	feed-down shape
generic $\Upsilon(4S) \rightarrow B\bar{B}$	DECAY.DEC	10^6	100–104/200–204	pseudo-data
continuum $q\bar{q}$	Pythia8	2.4×10^6	300–302	pseudo-data

The generic $B\bar{B}$ and continuum samples are combined into the *pseudo-dataset*. With $\sigma_{B\bar{B}} = 1.1\text{nb}$ and the measured generator cross section $\sigma_{q\bar{q}} = 2.63\text{nb}$, a luminosity-matched mixture requires $N_{q\bar{q}}/N_{B\bar{B}} = 2.39$; the residual mismatch of the generated 2.4×10^6 events is absorbed by a per-candidate weight of 0.995. The dataset corresponds to $N_{B\bar{B}}/\sigma_{B\bar{B}} \approx 0.9\text{fb}^{-1}$. The truth record counts $N_{B^0} = 965\,840$ neutral- B decays (mixing transitions excluded), of which 14 319 are true $B^0 \rightarrow D^{*-}\tau^+\nu_\tau$ — an effective generator branching fraction of 1.483%, the closure target.

3 Detector model and reconstruction

3.1 Parametric detector response

The detector response is deliberately simple — a handful of parameters in one configuration object — but each parameter is chosen with reference to the published Belle II performance [10, 11]:

Table 2: Fast-simulation parameters.

Effect	Model	Reference point
track momentum	$\sigma_p/p = 0.5\%$, direction unchanged, E recomputed from the true mass	typical for the $p \lesssim 2\text{GeV}$ tracks of these decays [10]
acceptance	$17^\circ < \theta_{\text{lab}} < 150^\circ$	CDC coverage [10]
tracking efficiency	95% per track, flat	conservative vs. published tracking
photons (ECL)	$\sigma_E/E = 2\%/\sqrt{E} \oplus 1\%$, $E > 50\text{MeV}$	simplified, slightly conservative form of the ECL resolution [10]
muon ID	$\varepsilon_\mu = 90\%$; $\pi \rightarrow \mu$ 2%, $K \rightarrow \mu$ 1%	magnitudes as in Belle II lepton-ID
electron ID	$\varepsilon_e = 95\%$; $\pi/K \rightarrow e$ 0.5%	performance [11]
mass hypothesis	identified leptons get the e/μ mass	

Deliberately *not* modeled at this stage, in roughly decreasing order of importance for this analysis: hadron-ID confusion ($K \leftrightarrow \pi$), electron bremsstrahlung, momentum- and angle-dependent ID rates, beam backgrounds, and material effects. A quantitative comparison of the resulting resolutions against public Belle II performance figures is in progress; the closure tests below establish internal consistency, not detector fidelity.

3.2 Candidate reconstruction

A candidate is a $D^{*\pm}$ whose stable charged decay products are exactly three tracks ($K^\mp \pi^\pm$ from the D^0 and a slow pion), paired with the highest- p^* opposite-charge track identified as a lepton — which, because of the fake rates above, may be a misidentified hadron. All four (or more) particles must pass the acceptance, efficiency, and ID response. One caveat is inherited from this truth-seeded construction: random $K\pi\pi$ combinations (fake D^*) are *not* modeled, so the Δm sideband is empty by construction (Sec. 5).

No true B information is used anywhere. All kinematic variables are computed in the e^+e^- CM frame with the beam constraint $p_B = (\sqrt{s}/2, \vec{0})$, so $B\bar{B}$ and continuum events are treated identically: $m_{\text{miss}}^2 = (p_B - p_{D^*} - p_\ell)^2$, $q^2 = (p_B - p_{D^*})^2$, p_ℓ^* , $\cos\theta_{BY}$, Fox–Wolfram R_2 , and the rest-of-event (ROE) variables built from every other detected particle. The ROE *energy* $E_{\text{ROE}}^{\text{CM}}$ is a powerful wrong-pairing discriminant (Sec. 4); the ROE *direction* carries no useful information at the $\Upsilon(4S)$ — the true B momentum (0.34 GeV) is overwhelmed by undetected particles (median direction error 58°), which is a quantitative illustration of why full tag reconstruction exists [12].

4 Selection optimization

4.1 Preselection: the D^* windows

The D^0 -mass and Δm windows are taken from the published analyses [3]: $|m(K\pi) - m_{D^0}| < 20$ MeV and $|\Delta m - 145.4 \text{ MeV}| < 2.5$ MeV. Figure 1 shows both distributions in the pseudo-dataset with the windows indicated; the narrow Δm peak is the classic slow-pion signature and is essentially background-free in this framework (see the fake- D^* caveat above).

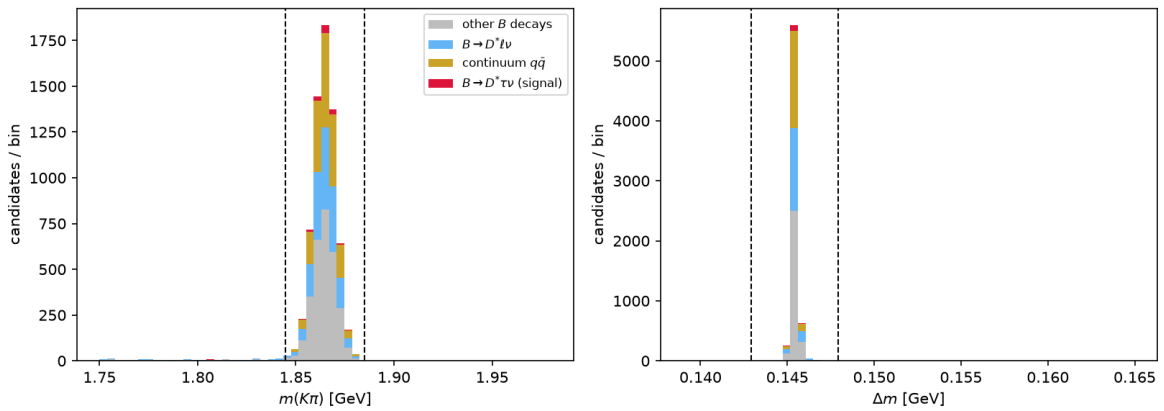


Figure 1: $m(K\pi)$ (left, after the Δm window) and Δm (right, after the $m(K\pi)$ window) in the pseudo-dataset, split by true origin. Dashed lines: selection windows.

4.2 Continuum suppression and wrong-pairing rejection

Two backgrounds dominate after the D^* windows, and each motivates one cut (Fig. 2):

- **Continuum $q\bar{q}$ events** are jet-like: $\langle R_2 \rangle = 0.39$ versus 0.16 for $B\bar{B}$. We require $R_2 < 0.3$.
- **Wrong lepton pairing** — a genuine D^* combined with a lepton from the other B or from a charm decay — is the single largest background. For a correctly paired candidate the ROE is exactly the tag B , so $E_{\text{ROE}}^{\text{CM}} \lesssim \sqrt{s}/2$ minus undetected energy; a wrongly paired candidate pulls signal-side particles into the ROE and pushes $E_{\text{ROE}}^{\text{CM}}$ up. We require $E_{\text{ROE}}^{\text{CM}} < 4.0$ GeV, which improves S/B in the signal region by a factor ≈ 4 at a signal cost visible in the cutflow (Table 4).

p_ℓ^* and $\cos\theta_{BY}$ were scanned as well and found to add no discrimination against the wrong-pairing background (both were kept unrestricted); this is expected, since wrongly paired leptons have nearly the same momentum spectrum as the soft signal leptons.

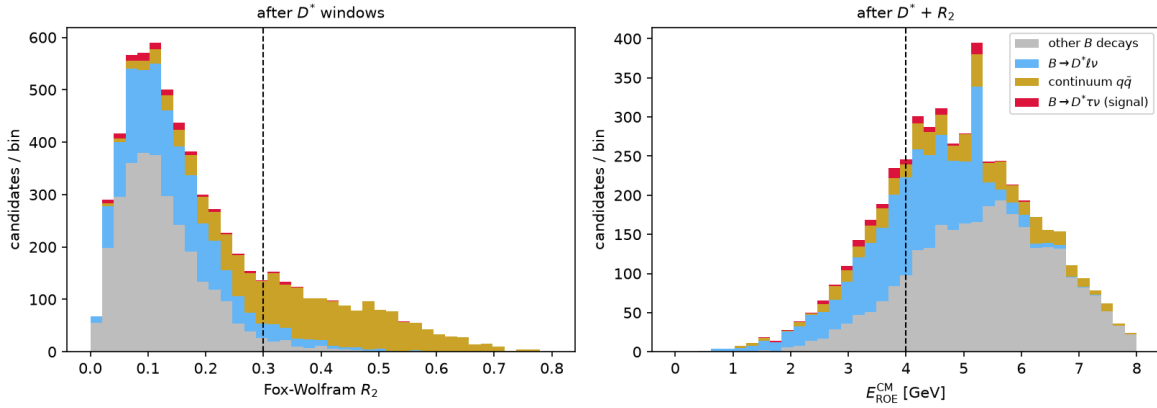


Figure 2: R_2 after the D^* windows (left) and $E_{\text{ROE}}^{\text{CM}}$ after the R_2 cut (right), split by true origin. Dashed lines: chosen cuts.

4.3 Signal region

The signal region is defined by $m_{\text{miss}}^2 > 1.5 \text{ GeV}^2$. The threshold scan on the full pseudo-dataset (Table 3) has a broad optimum — the figure of merit varies by $< 6\%$ over $1.0\text{--}2.0 \text{ GeV}^2$ — because the residual background is nearly flat in m_{miss}^2 while the $B^0 \rightarrow D^{*-}\ell^+\nu_\ell$ peak at zero is already far away; 1.5 GeV^2 sits at the plateau and suppresses the $B^0 \rightarrow D^{*-}\ell^+\nu_\ell$ leak to the sub-percent level (the normalization mode retains only $\sim 0.7\%$ of its candidates above the threshold).

5 Background estimation

5.1 Composition from truth matching

For this MC-only demonstration the background is decomposed with the truth labels carried by every candidate (true decay of the parent B , and the true species of the lepton candidate). Truth matching is what makes the generic $B\bar{B}$ sample usable as a background model at all: it contains *every* B decay mixed together, and the labels are the only way to know which component populates which region — and, importantly, to *validate the control regions we intend to use on data* (Sec. 5.2).

Table 3: m_{miss}^2 threshold scan on the pseudo-dataset (luminosity-weighted; $S = \text{true } B^0 \rightarrow D^{*-}\tau^+\nu_\tau$ with a true lepton, $B = \text{everything else}$).

$m_{\text{miss}}^2 > [\text{GeV}^2]$	S	B	$S/\sqrt{S+B}$
0.5	48	454	2.14
1.0	45	325	2.34
1.25	45	288	2.47
1.5	43	276	2.41
2.0	39	243	2.32
2.5	33	204	2.14

Table 4: Signal cutflow (forced signal MC, 5 000 generated events, both lepton channels; the reconstruction row includes acceptance, tracking efficiency, and lepton ID).

Stage	Candidates	step efficiency
reconstructed	2 801	56.0%
D^* mass windows	2 672	95.4%
$R_2 < 0.3$	2 482	92.9%
$E_{\text{ROE}}^{\text{CM}} < 4.0 \text{ GeV}$	1 282	51.7%
$m_{\text{miss}}^2 > 1.5 \text{ GeV}^2$	1 036	80.8%
total		$(20.7 \pm 0.6)\%$

Table 5: Composition of the 319 signal-region candidates (luminosity-weighted).

Component	Candidates	Fraction
signal (true $B^0 \rightarrow D^{*-}\tau^+\nu_\tau$, true lepton)	43	13%
other B decays, true lepton (wrong pairing)	194	61%
$B^0 \rightarrow D^{*-}\ell^+\nu_\ell$ resolution leak	48	15%
misidentified-hadron leptons	20	6%
continuum $q\bar{q}$	15	5%

5.2 Control regions and the strategy for data

With real data no truth labels exist, and the plan — following the standard practice of the published analyses [3, 5] — is to normalize each background component in a dedicated control region (CR) and extrapolate into the signal region with MC shapes, or equivalently to include the CRs as simultaneous channels in the fit. The pseudo-dataset lets us define these CRs now and measure their purity (Fig. 3):

- **wrong-pairing CR:** $E_{\text{ROE}}^{\text{CM}} > 5 \text{ GeV}$, $m_{\text{miss}}^2 > 1.5$ — 1584 candidates, **90% pure** in the wrong-pairing component, with 1% signal contamination. This is the CR that pins down the dominant background.
- $B^0 \rightarrow D^{*-}\ell^+\nu_\ell$ **CR:** $-2 < m_{\text{miss}}^2 < 0.5 \text{ GeV}^2$ after the full preselection — 621 candidates, **82% pure** $B^0 \rightarrow D^{*-}\ell^+\nu_\ell$. It normalizes the resolution leak and, on data, would directly validate the fast simulation’s m_{miss}^2 resolution model.
- **continuum CR:** $R_2 > 0.4$ — **82% pure** continuum (statistically limited here; on data, off-resonance running would replace it).

Not available in this framework: an $m(K\pi)/\Delta m$ sideband CR for fake- D^* combinatorics, because random $K\pi\pi$ combinations are not modeled (Sec. 3); on data this sideband is populated and provides the corresponding normalization.

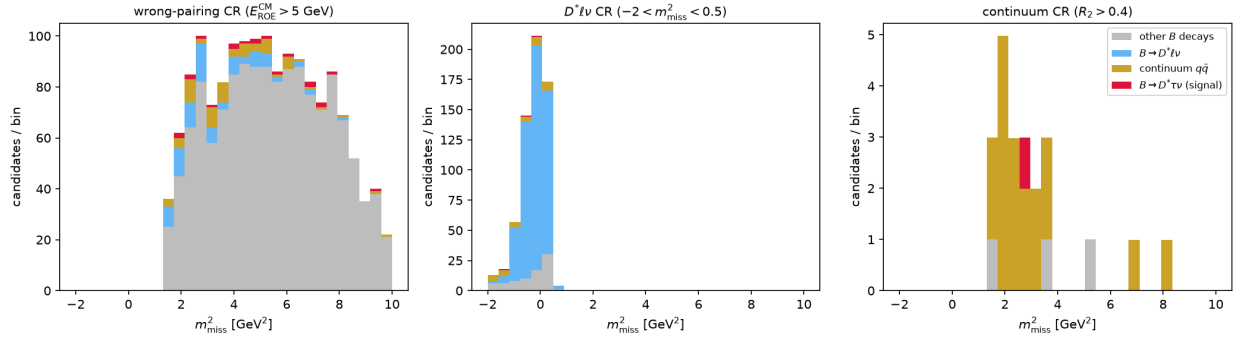


Figure 3: m_{miss}^2 in the three control regions, split by true origin.

6 Results

6.1 Cut and count

The branching fraction is extracted as

$$\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau) = \frac{N_{\text{obs}} - N_{\text{bkg}}}{N_{B^0} \varepsilon \mathcal{B}_{\text{sub}}}, \quad \mathcal{B}_{\text{sub}} = \mathcal{B}(D^* \rightarrow D^0\pi) \mathcal{B}(D^0 \rightarrow K\pi) \mathcal{B}(\tau \rightarrow \ell\nu\nu),$$

with $N_{B^0} = 965840$ from the truth record (on data: $2f^{00}N_{B\bar{B}}$), $\varepsilon = (20.7 \pm 0.6)\%$ from the forced signal sample, $\mathcal{B}_{\text{sub}} = 0.677 \times 0.0389 \times 0.331$ (decay-table values), $N_{\text{obs}} = 319$ and $N_{\text{bkg}} = 276$ from Table 5:

$$\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau) = (2.47 \pm 1.40_{\text{stat}})\%, \quad \text{truth } 1.483\% (+0.7\sigma).$$

Per lepton channel the signal region holds 153 candidates (21 true signal) for electrons and 166 (22) for muons — flavor-symmetric within statistics, as expected.

6.2 Lepton-flavor treatment

Electrons and muons are reconstructed as separate channels (they have different identification efficiencies, fake rates, and radiative tails) and *combined at the extraction stage* — the practice of the published analyses: the untagged $|V_{cb}|$ measurement quotes per-flavor results and averages them [3], while the $R(D^*)$ measurements fit the e and μ subsamples as simultaneous categories sharing the signal strength [5, 6]. We follow the latter: the fit below has one e and one μ channel with a common μ_{sig} , and the counting analysis reports per-flavor yields (153/166 candidates, 21/22 true signal in e/μ).

6.3 Binned template fit

The counting analysis discards the shape information; the statistical analysis is therefore a binned maximum-likelihood template fit with `pyhf` [13]. In each lepton channel the m_{miss}^2 distribution after the full preselection (no m_{miss}^2 cut) is histogrammed in 24 bins over $[-2, 10] \text{ GeV}^2$ and modeled as

$$\lambda_b = \mu S_b + \gamma_b B_b,$$

where μ is the signal strength (the parameter of interest, $\mathcal{B} = \mu \times 1.483\%$), S_b and B_b are the signal and background templates, and γ_b are per-bin nuisance parameters constrained to unity with width $\sigma_b = \sqrt{\sigma_{\text{MC},b}^2 + (0.10 B_b)^2}$ — the MC statistical uncertainty combined with a conservative 10% normalization placeholder for the not-yet-evaluated background systematics. The likelihood is the product of Poisson terms over bins times the constraint terms; it is maximized with MINUIT. The fit gains over counting because the $B^0 \rightarrow D^{*-}\ell^+\nu_\ell$ peak and the sidebands constrain the background normalization in situ:

$$\mu = 1.000 \pm 0.538 \quad \Rightarrow \quad \mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau) = (1.48 \pm 0.80)\%.$$

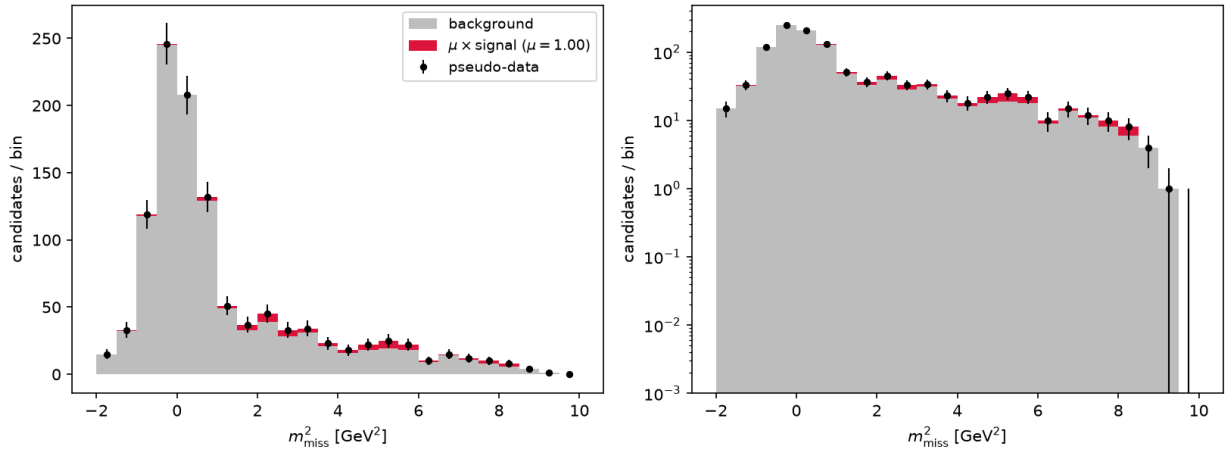


Figure 4: Post-fit m_{miss}^2 distribution in linear (left) and logarithmic (right) scale.

6.4 Fit validation and method comparison

The fit is validated with 300 toy experiments: pseudo-data are drawn as Poisson fluctuations of the $\mu = 1$ expectation (including the auxiliary measurements of the constraint terms) and refit. The

pull distribution $(\hat{\mu} - 1)/\sigma_{\hat{\mu}}$ has mean $+0.03$ and width 0.97 (Fig. 5): the estimator is unbiased and the reported uncertainty is correct.

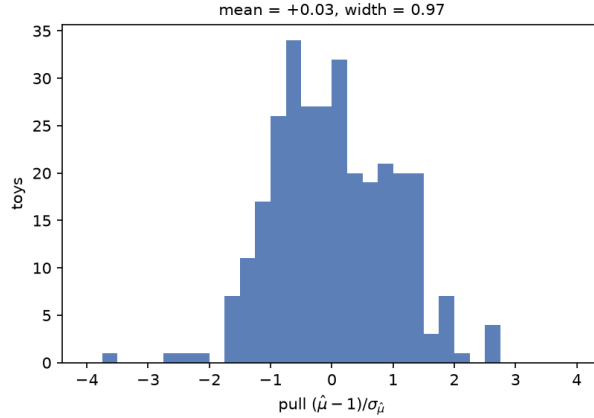


Figure 5: Pull distribution of $\hat{\mu}$ over 300 toy experiments.

Table 6: Method comparison on the same pseudo-dataset. The ROE row shows the impact of the wrong-pairing rejection; “ μ only” shows the gain from adding the electron channel.

Method	$\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau)$	rel. stat. @ 1 ab^{-1}
cut & count, no $E_{\text{ROE}}^{\text{CM}}$ cut	$(2.77 \pm 2.19)\%$	$\sim 4.5\%$
cut & count	$(2.47 \pm 1.40)\%$	$\sim 2.8\%$
template fit, μ channel only	$(1.48 \pm 0.89)\%$	$\sim 1.9\%$
template fit, $e + \mu$ merged	$(1.48 \pm 0.80)\%$	$\sim 1.7\%$
template fit, $e + \mu$ simultaneous	$(1.48 \pm 0.69)\%$	$\sim 1.4\%$

An *unbinned* likelihood fit is not yet implemented; the ntuples retain the per-candidate information needed, and it is a planned upgrade. Note that the fit’s central value closes exactly by construction — the signal template is currently the truth-labeled component of the pseudo-dataset itself; the meaningful outputs are the uncertainty and the toy validation. Refitting with the template taken from the independent forced signal sample is a planned refinement.

7 Discussion

Closure holds at every stage of the framework’s development — truth level, beam-constrained reconstruction, with continuum added, with lepton fake rates added — which we regard as the central validation: the framework returns the input branching fraction with correctly estimated, toy-validated uncertainties.

The known limitations, in rough order of importance:

1. **Truth-based background subtraction.** The data strategy of Sec. 5.2 (CR normalization or a simultaneous CR+SR fit) is designed and its CRs are measured to be 82–90% pure, but it has not yet replaced the truth subtraction in the pipeline.
2. **Systematic uncertainties are not yet propagated.** As a scale reference, typical magnitudes in comparable Belle (II) measurements are: tracking efficiency $\sim 0.3\%$ /track ($\sim 1.4\%$

for four tracks), lepton identification 1–2%, $\mathcal{B}(D^* \rightarrow D^0\pi)$ 0.7%, $\mathcal{B}(D^0 \rightarrow K\pi)$ 0.8%, $N_{B\bar{B}}$ and f^{00} 1–2% [3, 6, 14]. Added in quadrature this suggests a systematic floor of roughly 3%, comparable to the 1 ab^{-1} statistical precision — i.e. the measurement becomes systematics-limited exactly where a real analysis would, and a faithful systematics model (adopting these published values as priors, and varying the fast-simulation parameters for the detector-model part) is the top-priority next step.

3. **Detector-model fidelity.** Parametric response, constant ID rates, no bremsstrahlung, no fake- D^* combinatorics; a systematic comparison against public Belle II performance figures is in progress.
4. **Form factors.** ISGW2 throughout; migration to BGL/CLN is planned and will matter at the few-percent level for acceptances.
5. **Self-closure of the fit template** (see Sec. 6).

A second demonstration on the fully reconstructed rare decay $B^0 \rightarrow K^{*0}\ell^+\ell^-$ ($M_{bc}/\Delta E$ selection, charmonium vetoes) has been completed with the same pipeline and will be documented in a separate note.

8 Conclusion

An LLM agent operating under human review, over an entirely open-source, laptop-scale pipeline, carried a branching-fraction analysis from event generation through selection optimization, background decomposition with validated control regions, a toy-validated `pyhf` template fit, and the analysis note itself. On a 0.9 fb^{-1} -equivalent pseudo-dataset it recovers the generator-truth $\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau) = 1.48\%$ as $(1.48 \pm 0.80)\%$, corresponding to $\sim 1.7\%$ relative statistical precision at 1 ab^{-1} .

This note is meant as a concrete illustration for colleagues of what the approach produces. We would welcome thoughts on whether and how such a workflow could eventually be exercised on official samples or future open data — we understand that is a discussion for the appropriate bodies, and nothing in the present study depends on it.

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